Introduction to Probability and Statistics Fourteenth Edition

Chapter 11
The Analysis of Variance (ANOVA)

Variance vs Variation

- Suppose we have data $x_1, \ldots, x_n$.
- The sample variance: $\dfrac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$
- The variation for the data: $\sum_{i=1}^{n}(x_i - \bar{x})^2$

11.1 The design of an experiment

Extensions to Chapter 10

Two Population Means	Multiple Population Means
10.3: Indepedence t test	11.2 The Completely Radomized Block Design
10. 4 Paired t test	11.4 The Randomized Block Design

Experimental Design

- The **sampling plan** or **experimental design** determines the way that a sample is selected.
- In an **observational study,** the experimenter observes data that already exist. The **sampling plan** is a plan for collecting this data.
- In a **designed experiment,** the experimenter imposes one or more experimental conditions on the experimental units and records the response.

Definitions

- An **experimental unit** is the object on which a measurement or measurements) is taken.
- A **factor** is an independent variable whose values are controlled and varied by the experimenter. (Also known as explaintory variable.)
- A **level** is the intensity setting of a factor.
- A **treatment** is a specific combination of factor levels.
- The **response** is the variable being measured by the experimenter.

EXAMPLE

- A group of people is randomly divided into an experimental and a control group. The control group is given an aptitude test after having eaten a full breakfast. The experimental group is given the same test without having eaten any breakfast.

Experimental unit = **person**

Factor = **meal**

Response = **Score on test**

Levels = **Breakfast** or **no breakfast**

Treatments: **Breakfast** or **no breakfast**

EXAMPLE

- The experimenter in the previous example also records the person's gender. Describe the factors, levels and treatments.

Experimental unit = **person**

Response = **score**

Factor #1 = **meal**

Factor #2 = **gender**

Levels = **breakfast** or **no breakfast**

Levels = **male** or **female**

Treatments:

male and **breakfast, female** and **breakfast, male** and **no breakfast, female** and **no breakfast**

The Analysis of Variance (ANOVA)

- All measurements exhibit **variability**.
- The total variation in the response measurements is broken into portions that can be attributed to various **factors**.
- These portions are used to judge the effect of the various factors on the experimental response.

The Analysis of Variance

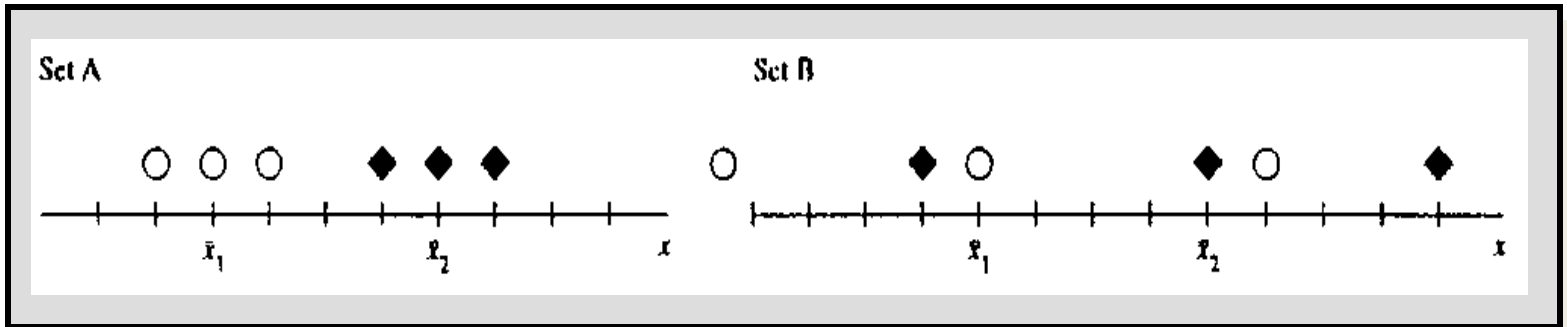
- If an experiment has been properly designed,

Total variation → **Factor 1**, **Factor 2**, **Random variation**

•We compare the variation due to any one factor to the typical random variation in the experiment.

The variation between the sample means is larger than the typical variation within the samples.

The variation between the sample means is about the same as the typical variation within the samples.



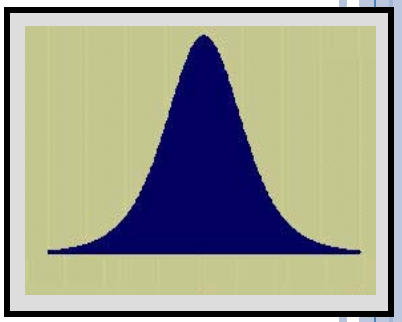
Assumptions

- Similar to the assumptions required in Chapter 10.

1. The observations within each population are normally distributed with a common variance σ^2.
2. Assumptions regarding the sampling procedures are specified for each design.

•Analysis of variance procedures are fairly robust when sample sizes are equal and when the data are fairly mound-shaped.

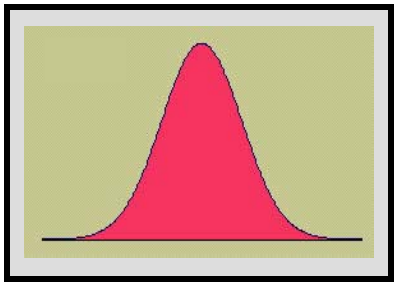
Three Designs



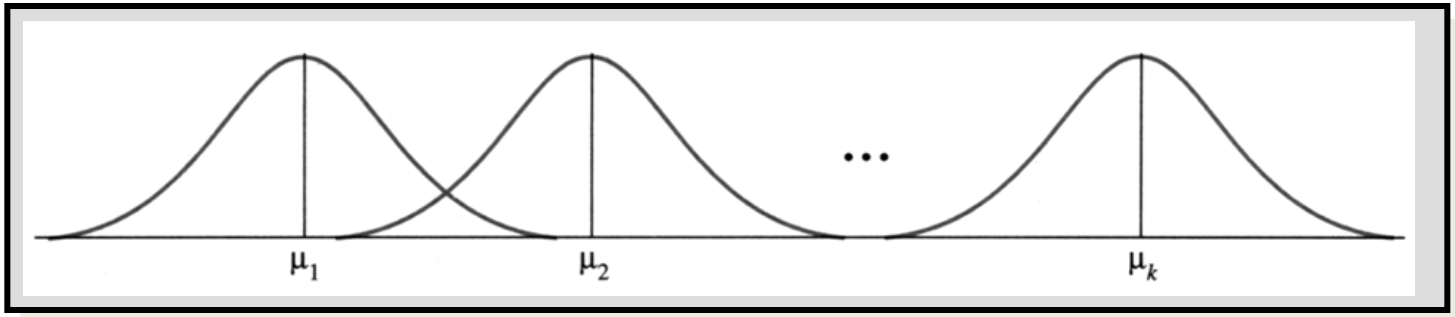
- **Completely randomized design:** an extension of the two independent sample t-test.
- **Randomized block design:** an extension of the paired difference test.
- **a × b Factorial experiment:** we study two experimental factors and their effect on the response.

11.2 The completely randomized design: a one-way classification

The Completely Randomized Design



- A **one-way classification** in which one factor is set at k different levels.
- The k levels correspond to k different normal populations, which are the **treatments**.
- Are the k population means the same, or is at least one mean different from the others?



EXAMPLE

Is the attention span of children affected by whether or not they had a good breakfast? Twelve children were randomly divided into three groups and assigned to a different meal plan. The response was attention span in minutes during the

No Breakfast	Light Breakfast	Full Breakfast
8	14	10
7	16	12
9	12	16
13	17	15

k = 3 treatments. Are the average attention spans different?

The Completely Randomized Design

- Random samples of size n_1, n_2, …,n_k are drawn from k populations with means μ_1, μ_2,…, μ_k and with common variance σ^2.
- Let x_{ij} be the j-th measurement in the i-th sample.
- The total variation in the experiment is measured by the **total sum of squares**:

$$\text{Total SS} = \sum (x_{ij} - \bar{x})^2$$

The completely randomized design

- Random samples of size $n_1, n_2, \ldots, n_k$ are drawn from k populations with means $\mu_1, \ldots, \mu_k$ and with common variance σ^2.
- Let x_{ij} be the -th measurement in the i-th sample.

- Model:
- $$x_{ij} \sim \mathcal{N}(\mu_i, \sigma^2),$$
- for $i = 1, \ldots, k$, $j = 1, \ldots, n_i$, with total sample size $n_1 + \cdots + n_k = n$.

Hypothesis Test

1. $H_0 : \mu_1 = \cdots = \mu_k$ vs H_a: at least one mean is different.
2. $\alpha = 0.05$
3. Test statistic and its distribution:

$$F_{\text{STAT}} = \frac{\text{MST}}{\text{MSE}} \sim F_{(k-1),(n-k)} \longleftarrow \boxed{\textbf{ANOVA table!}}$$

4. Calculate the realized F^*.
5. This is a right-tailed test.
   - The rejection region is: $\left\{ F : F > F_{(k-1),(n-k),\alpha} \right\}$
   - The p-value is: $P\left(F_{(k-1),(n-k)} > F^* \right)$
6. Conclude

The total sum of squares, total SS

The grand mean $\bar{G} = \frac{1}{n}\sum_{ij} x_{ij}$.

$$\text{Total SS} = \sum_{i,j}(x_{ij} - \bar{G})^2.$$

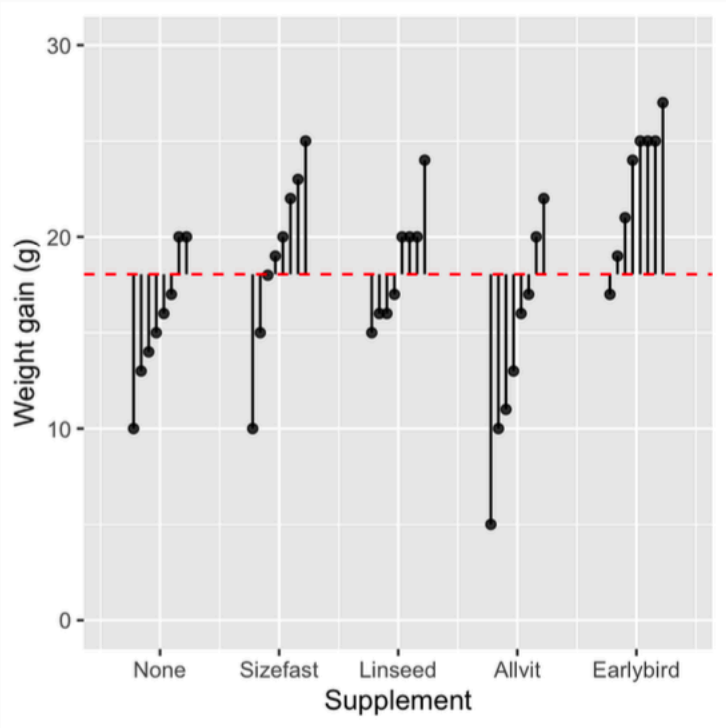
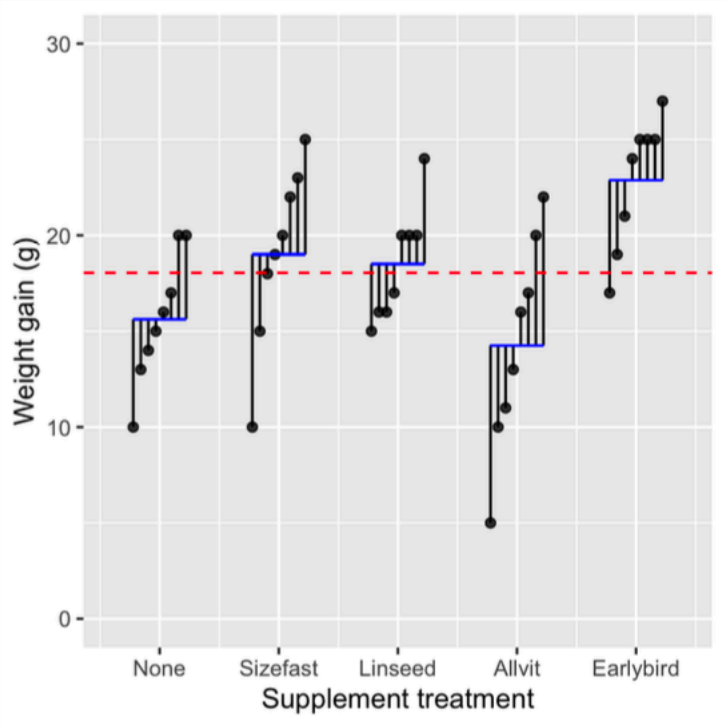


Photo credit: https://dzchilds.github.io/stats-for-bio

Sum squares of treatment, SST

The mean for treatment i: $\bar{T}_i = \frac{1}{n_i} \sum_{j=1,\ldots,n_i} x_{ij}$.

$$\text{SST} = \sum_i n_i (\bar{T}_i - \bar{G})^2$$



ANOVA table

$$\textbf{SSE} = \sum_{ij} (x_{ij} - \bar{T}_i)^2 = \textbf{Total SS - SST}$$

Soure	SS	df	MS	F
Treatment	SST	$(k-1)$	MST = (SST/$(k-1)$)	$F = \frac{MST}{MSE}$
Error	SSE	$(n-k)$	MSE = (SSE/$(n-k)$)	
Total	Total SS	$(n-1)$		

Degrees of freedom and mean squares

These sums of squares behave like the numerator of a sample variance.

When divided by the appropriate degrees of freedom, each provides a mean square, an estimate of variation in the experiment.

Degrees of freedom are additive, just like the sums of squares.

The Analysis of Variance

The **Total SS** is divided into two parts:

- ✓ **SST** (sum of squares for treatments): measures the variation among the k sample means.
- ✓ **SSE** (sum of squares for error): measures the variation within the k samples.

in such a way that:

$$\text{Total SS} = \text{SST} + \text{SSE}$$

Computing Formulas

$$\text{Total SS} = \Sigma x_{ij}^2 - \text{CM}$$
$$= (\text{Sum of squares of all } x\text{-values}) - \text{CM}$$

SOP

with

$$\text{CM} = \frac{(\Sigma x_{ij})^2}{n} = \frac{G^2}{n}$$

$$\text{SST} = \Sigma \frac{T_i^2}{n_i} - \text{CM}$$

$$\text{SSE} = \text{Total SS} - \text{SST}$$

and

G = Grand total of all n observations

T_i = Total of all observations in sample i

n_i = Number of observations in sample i

$n = n_1 + n_2 + \cdots + n_k$

1. $G = \sum_{ij} x_{ij}$
2. $\text{CM} = G^2/n$
3. $T_i = \sum_{j=1,\ldots,n_i} x_{ij}$
4. $\text{Total SS} = \sum_{ij} x_{ij}^2 - CM$
5. $\text{SST} = \sum \frac{T_i^2}{n_i} - CM$
6. $\text{SSE} = \text{Total SS} - \text{SST}$.

CM=correction for the mean

Why?

Simplified computing formulas for the ANOVA table I

Define

$$G = \sum x_{ij} = \text{grand total of all } n \text{ observations},$$

$$T_i = \sum_{j}^{n_i} x_{ij} = \text{Total of all observations in sample } i,$$

$$n_i = \text{Number of observations in sample } i,$$

$$n = n_1 + n_2 + \cdots + n_k.$$

Second, define

$$CM = \frac{1}{n} G^2.$$

Simplified computing formulas for the ANOVA table II

First, we have

$$CM = \frac{G^2}{n} = n\left(\frac{G}{n}\right)^2 = n\bar{G}^2$$

Second, recall that

$$\bar{T}_i = \frac{1}{n_i} T_i.$$

Taking squares of both sides, we have

$$\bar{T}_i^2 = \frac{1}{n_i^2} T_i^2,$$

and hence

$$n_i \bar{T}_i^2 = \frac{1}{n_i} T_i^2.$$

Simplified computing formulas for the ANOVA table III

With $\sum x_{ij} = n\bar{G}$, we can calculate

Original definition

$$
\begin{aligned}
\text{Total SS} &= \sum (x_{ij} - \bar{G})^2 \\
&= \sum (x_{ij}^2 - 2\bar{G}x_{ij} + \bar{G}^2) \\
&= \sum x_{ij}^2 - 2\bar{G}(n\bar{G}) + n\bar{G}^2 \\
&= \sum x_{ij}^2 - n\bar{G}^2 \\
&= \sum x_{ij}^2 - CM.
\end{aligned}
$$

Used for hand calculation

Simplified computing formulas for the ANOVA table IV

Similarly, we have

Original definition

$$
\begin{aligned}
\text{SST} &= \sum_i n_i(\bar{T}_i - \bar{G})^2 \\
&= \sum_i n_i(\bar{T}_i - 2\bar{T}_i\bar{G} + \bar{G}^2) \\
&= \sum_i (n_i\bar{T}_i - 2n_i\bar{T}_i\bar{G} + n_i\bar{G}^2) \\
&= \sum_i \frac{T_i^2}{n_i} - 2(n\bar{G})\bar{G} + n\bar{G}^2 \\
&= \sum_i \frac{T_i^2}{n_i} - n\bar{G}^2 \\
&= \sum_i \frac{T_i^2}{n_i} - CM.
\end{aligned}
$$

Used for hand calculation

The Breakfast Problem

No Breakfast	Light Breakfast	Full Breakfast
8	14	10
7	16	12
9	12	16
13	17	15
$T_1 = 37$	$T_2 = 59$	$T_3 = 53$

G = 149

$$CM = \frac{149^2}{12} = 1850.0833$$

$$\text{Total SS} = 8^2 + 7^2 + \ldots + 15^2 - CM = 1973 - 1850.0833 = 122.9167$$

$$SST = \frac{37^2}{4} + \frac{53^2}{4} + \frac{59^2}{4} - CM = 1914.75 - CM = 64.6667$$

$$SSE = \text{Total SS} - SST = 58.25$$

Calculation using numbers/excels

x_{ij}	No breakfast	Light breakfast	Full breakfast		x_{ij}^2	No breakf	Light brea	Full break
	8	14	10			64	196	100
	7	16	12			49	256	144
	9	12	16			81	144	256
	13	17	15			169	289	225
T_i	37	59	53					
T_i^2	1369	3481	2809					
G	149							
CM	1850.0833							
	SS	df	MSE	F				
SST	64.6667	2	32.3333	4.9957				
SSE	58.2500	9	6.4722					
total SS	122.9167	11						

SOP

1. $G = \sum_{ij} x_{ij}$
2. $\text{CM} = G^2/n$
3. $T_i = \sum_{j=1,\ldots,n_i} x_{ij}$
4. $\text{Total SS} = \sum_{ij} x_{ij}^2 - CM$
5. $\text{SST} = \sum \frac{T_i^2}{n_i} - CM$
6. $\text{SSE} = \text{Total SS} - \text{SST}.$

Degrees of Freedom and Mean Squares

- These **sums of squares** behave like the numerator of a sample variance. When divided by the appropriate **degrees of freedom**, each provides a **mean square**, an estimate of variation in the experiment.
- **Degrees of freedom** are additive, just like the sums of squares.

$$\text{Total } df = \text{Trt } df + \text{Error } df$$

The ANOVA Table

Total df = $n_1+n_2+\ldots+n_k-1 = n-1$

Treatment df = $k-1$

Error df = $n-1-(k-1) = n-k$

Mean Squares

$MST = SST/(k-1)$

$MSE = SSE/(n-k)$

Source	df	SS	MS	F
Treatments	k -1	SST	MST= SST/(k-1)	MST/MSE
Error	n - k	SSE	MSE=SSE/(n-k)	
Total	n -1	Total SS		

The Breakfast Problem

$$CM = \frac{149^2}{12} = 1850.0833$$

$$\text{Total SS} = 8^2 + 7^2 + \ldots + 15^2 - CM = 1973 - 1850.0833 = 122.9167$$

$$SST = \frac{37^2}{4} + \frac{53^2}{4} + \frac{59^2}{4} - CM = 1914.75 - CM = 64.6667$$

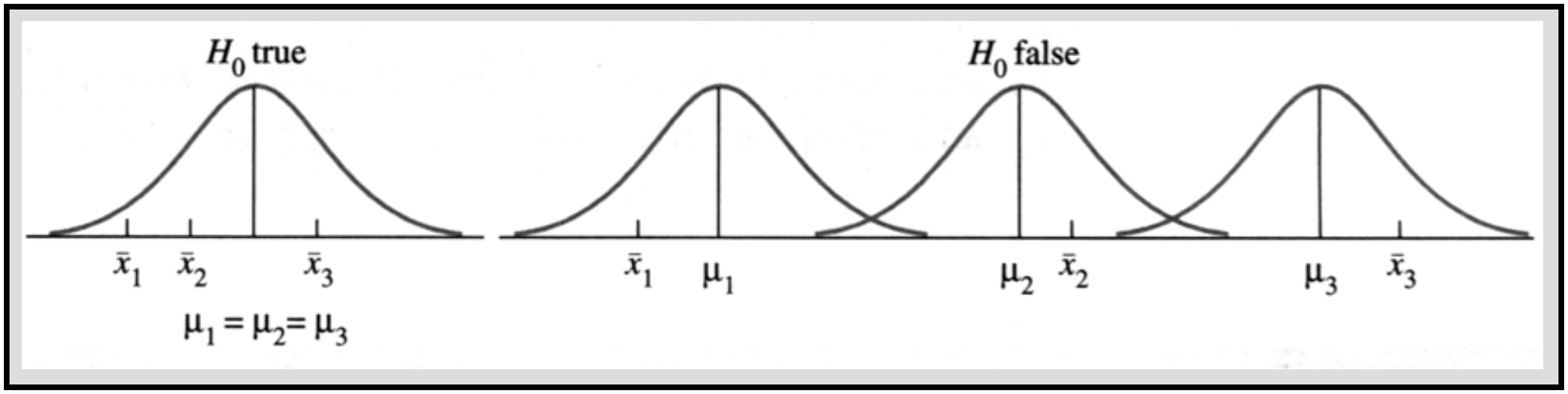
$$SSE = \text{Total SS} - SST = 58.25$$

Source	df	SS	MS	F
Treatments	2	64.6667	32.3333	5.00
Error	9	58.25	6.4722	
Total	11	122.9167		

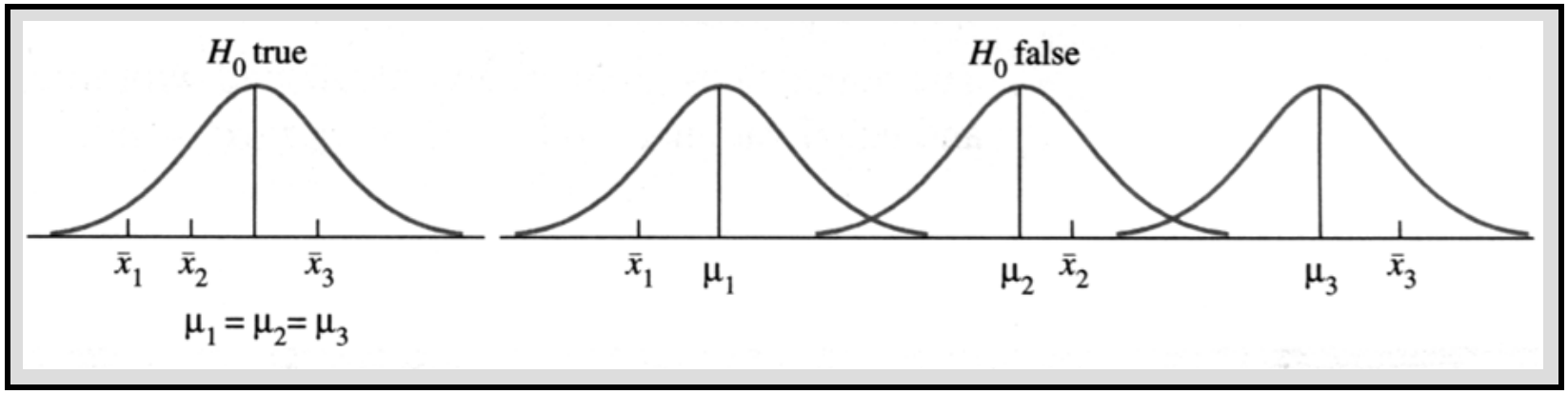
Testing the Treatment Means

$H_0 : \mu_1 = \mu_2 = \mu_3 = \ldots = \mu_k$ versus

H_a : at least one mean is different

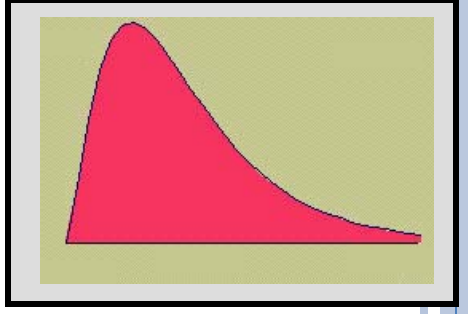


Remember that σ^2 is the common variance for all k populations. The quantity MSE = SSE/($n - k$) is a pooled estimate of σ^2, a weighted average of all k sample variances, whether or not H_0 is true.



- If H_0 is true, then the variation in the sample means, measured by MST = [SST/ (k – 1)], also provides an unbiased estimate of σ^2.
- However, if H_0 is false and the population means are different, then MST— which measures the variance in the sample means — is unusually **large.** The test statistic **F = MST/ MSE** tends to be larger that usual.

MY APPLET



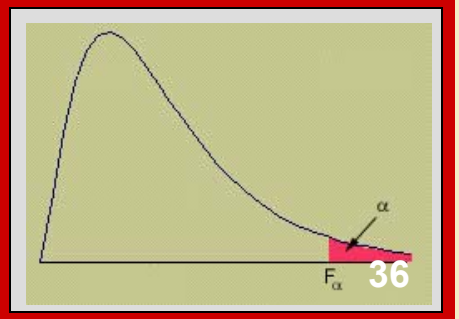
THE F TEST

- Hence, you can reject H 0 for large values of F, using a right-tailed statistical test.
- When H_0 is true, this test statistic has an F distribution with $df_1 = (k - 1)$ and $df_2 = (n - k)$ degrees of freedom and **right-tailed** critical values of the F distribution can be used.

To test $H_0 : \mu_1 = \mu_2 = \mu_3 = \ldots = \mu_k$

Test Statistic: $F = \dfrac{MST}{MSE}$

Reject H_0 if $F > F_\alpha$ with $k - 1$ and n-k df.



The Breakfast Problem

Source	df	SS	MS	F
Treatments	2	64.6667	32.3333	5.00
Error	9	58.25	6.4722	
Total	11	122.9167		

$H_0 : \mu_1 = \mu_2 = \mu_3$ versus

H_a : at least one mean is different

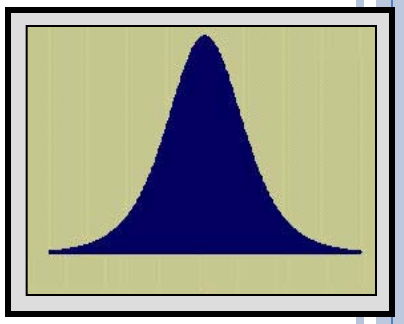
$$F = \frac{MST}{MSE} = \frac{32.3333}{6.4722} = 5.00$$

Rejection region : $F > F_{.05} = 4.26$.

We reject H_0 and conclude that there is a difference in average attention spans.

MY APPLET

Confidence Intervals



- If a difference exists between the treatment means, we can explore it with confidence intervals.

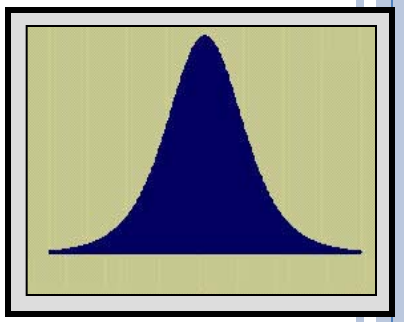
A single mean, $\mu_i : \bar{x}_i \pm t_{\alpha/2} \dfrac{s}{\sqrt{n_i}}$

Difference $\mu_i - \mu_j : (\bar{x}_i - \bar{x}_j) \pm t_{\alpha/2} \sqrt{s^2 \left(\dfrac{1}{n_i} + \dfrac{1}{n_j} \right)}$

where $s = \sqrt{\text{MSE}}$ and t is based on error df.

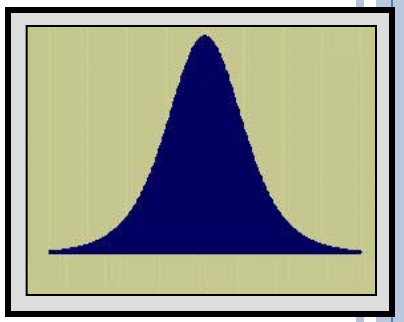
11.3 Ranking population means

Tukey's Method for Paired Comparisons



- Designed to test all pairs of population means simultaneously, with an **overall error rate of** α (ie., the familywise error rate, FWER).
- Based on the **studentized range**, the difference between the largest and smallest of the *k* sample means.
- Assume that the **sample sizes are equal** (n_i) and calculate a "ruler" that measures the distance required between any pair of means to declare a significant difference.

Tukey's Method



Calculate: $\omega = q_\alpha(k, df)\dfrac{s}{\sqrt{n_i}}$

where k = number of treatment means

$s = \sqrt{\text{MSE}}$ $\qquad df$ = error df

n_i = common sample size

$q_\alpha(k, df)$ = value from Table 11.

If any pair of means differ by more than ω, they are declared different.

Table 11(a) Percentage Points of the Studentized Range, $q_{.05}(k, df)$; Upper 5% Points

	k									
df	2	3	4	5	6	7	8	9	10	11
1	17.97	26.98	32.82	37.08	40.41	43.12	45.40	47.36	49.07	50.59
2	6.08	8.33	9.80	10.88	11.74	12.44	13.03	13.54	13.99	14.39
3	4.50	5.91	6.82	7.50	8.04	8.48	8.85	9.18	9.46	9.72
4	3.93	5.04	5.76	6.29	6.71	7.05	7.35	7.60	7.83	8.03
5	3.64	4.60	5.22	5.67	6.03	6.33	6.58	6.80	6.99	7.17
6	3.46	4.34	4.90	5.30	5.63	5.90	6.12	6.32	6.49	6.65
7	3.34	4.16	4.68	5.06	5.36	5.61	5.82	6.00	6.16	6.30
8	3.26	4.04	4.53	4.89	5.17	5.40	5.60	5.77	5.92	6.05
9	3.20	3.95	4.41	4.76	5.02	5.24	5.43	5.59	5.74	5.87
10	3.15	3.88	4.33	4.65	4.91	5.12	5.30	5.46	5.60	5.72
11	3.11	3.82	4.26	4.57	4.82	5.03	5.20	5.35	5.49	5.61
12	3.08	3.77	4.20	4.51	4.75	4.95	5.12	5.27	5.39	5.51
13	3.06	3.73	4.15	4.45	4.69	4.88	5.05	5.19	5.32	5.43
14	3.03	3.70	4.11	4.41	4.64	4.83	4.99	5.13	5.25	5.36
15	3.01	3.67	4.08	4.37	4.60	4.78	4.94	5.08	5.20	5.31
16	3.00	3.65	4.05	4.33	4.56	4.74	4.90	5.03	5.15	5.26
17	2.98	3.63	4.02	4.30	4.52	4.70	4.86	4.99	5.11	5.21
18	2.97	3.61	4.00	4.28	4.49	4.67	4.82	4.96	5.07	5.17
19	2.96	3.59	3.98	4.25	4.47	4.65	4.79	4.92	5.04	5.14
20	2.95	3.58	3.96	4.23	4.45	4.62	4.77	4.90	5.01	5.11
24	2.92	3.53	3.90	4.17	4.37	4.54	4.68	4.81	4.92	5.01
30	2.89	3.49	3.85	4.10	4.30	4.46	4.60	4.72	4.82	4.92
40	2.86	3.44	3.79	4.04	4.23	4.39	4.52	4.63	4.73	4.82
60	2.83	3.40	3.74	3.98	4.16	4.31	4.44	4.55	4.65	4.73
120	2.80	3.36	3.68	3.92	4.10	4.24	4.36	4.47	4.56	4.64
∞	2.77	3.31	3.63	3.86	4.03	4.17	4.29	4.39	4.47	4.55

The Breakfast Problem

Use Tukey's method to determine which of the three population means differ from the others.

	No Breakfast	Light Breakfast	Full Breakfast
	$T_1 = 37$	$T_2 = 59$	$T_3 = 53$
Means	37/4 = 9.25	59/4 = 14.75	53/4 = 13.25

Percentage Points of the Studentized Range, $q(k, df)$; Upper 5% Points

df	k = 2	3	4	5	6
1	17.97	26.98	32.82	37.08	40.41
2	6.08	8.33	9.80	10.88	11.74
3	4.50	5.91	6.82	7.50	8.04
4	3.93	5.04	5.76	6.29	6.71
5	3.64	4.60	5.22	5.67	6.03
6	3.46	4.34	4.90	5.30	5.63
7	3.34	4.16	4.68	5.06	5.36
8	3.26	4.04	4.53	4.89	5.17
9	3.20	3.95	4.41	4.76	5.02

$$\omega = q_{.05}(3,9)\frac{s}{\sqrt{4}} = 3.95\frac{\sqrt{6.4722}}{\sqrt{4}} = 5.02$$

The Breakfast Problem

List the sample means from smallest to largest.

$\bar{x}_1$	$\bar{x}_3$	$\bar{x}_2$
9.25	13.25	14.75

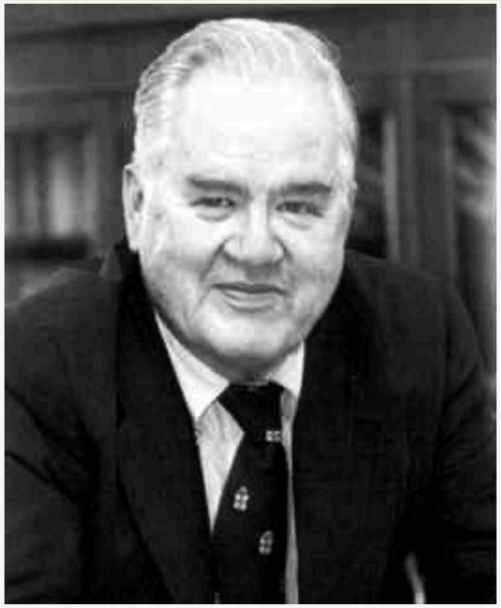
$\omega = 5.02$

We can declare a significant difference in average attention spans between "no breakfast" and "light breakfast", but not between the other pairs.

- Since the difference between 9.25 and 13.25 is less than ω = 5.02, there is no significant difference between population mean 1, μ_1 and population mean 3, μ_3.
- Since the difference between 13.25 and 14.75 is less than ω = 5.02, there is no significant difference between population mean 3, μ_3, and population mean 2, μ_2.
- Since the difference between 9.25 and 14.75 is larger than ω = 5.02, there is a significant difference between population mean 1, μ_1, and, population mean 2, μ_2 however.

More on Tukey's method

- Cooley-Tukey FFT algoritm
- Boxplot in his 1977 book, "Exploratory Data Analysis".
- Tukey's range test
  - Tukey's test
  - Tukey method
  - Tukey's honest significance test
  - Tukey's HSD (honestly signifiant difference) test



John Wilder Tukey

1915-2020. Wiki

https://en.wikipedia.org/wiki/Tukey's_range_test

Multiple comparisons

- In completely randomized design
  $H_0 : \mu_1 = \mu_2 = \cdots = \mu_k$,
- Simultaneously need $C_2^k = k(k-1)/2$ tests
  - $H_0 : \mu_1 = \mu_2$
  - $H_0 : \mu_1 = \mu_3$
  - $\vdots$
  - $H_0 : \mu_{k-1} = \mu_k$

	H0 true	H0 false
Reject H0	Type I error α	
Don't reject H0	$1-\alpha$	Type II error

Type I error inflation

- Type I error = false positive = false discovery

  $\alpha = \Pr(\text{Type I error}) = \Pr(\text{reject } H_0 \mid H_0 \text{ true})$
- Suppose you use a two-sample t test in a multiple comparisons.
- Set $\alpha = 0.05$ in a hypothesis test.
  - Do just 1 test:
    $\alpha_{new} = 1 - (1 - \alpha) = 1 - 0.95 = 0.05$
  - Do 2 tests:
    $\alpha_{new} = 1 - (1 - \alpha)^2 = 1 - 0.95^2 = 0.0975$
  - Do 6 tests:
    $\alpha_{new} = 1 - (1 - \alpha)^6 = 1 - 0.95^6 = 0.2649$

Family-wise error rate

- The FWER

$= \Pr(\text{making one or more false discoveries})$

$= \Pr(\text{Type I errors when performing multiple tests})$.

- Let V be the number of false positives in multiple comparisons (N):

$$\text{FWER} = \Pr(V \geq 1) = 1 - \Pr(V = 0) = 1 - (1 - \alpha)^N .$$

- Tukey's method: make $\text{FWER} \leq \alpha$

The Studentized range (q) distribution

- The studentized range distribution

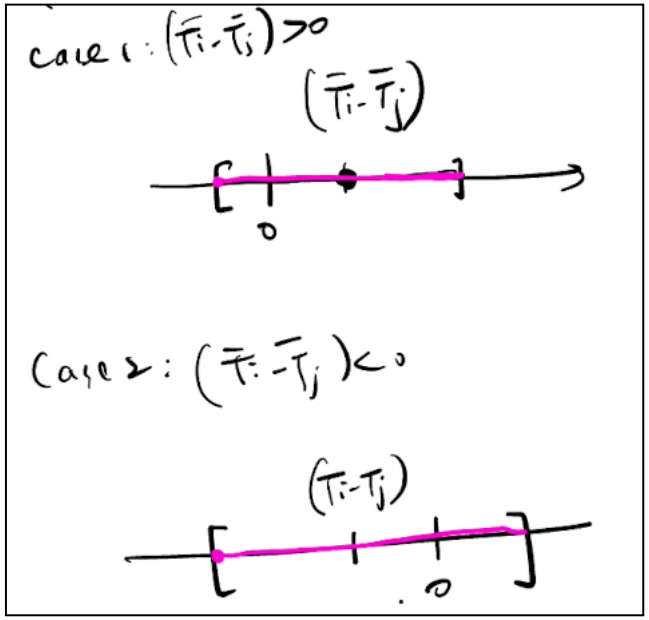
$$q_{STAT} = \frac{(\bar{T}_{\max} - \bar{T}_{\min}) - (\mu_i - \mu_j)}{s/\sqrt{n_i}} \sim q_{k,(n-k)}$$

k: # treatments, n: # observations, $\bar{T}_i$: treatment means, n_i is the common sample size of each treatment.

- The interval estimate for $(\mu_i - \mu_j)$ is

$$(\bar{T}_i - \bar{T}_j) \pm q_\alpha(k, n-k)\frac{s}{\sqrt{n_i}} = (\bar{T}_i - \bar{T}_j) \pm \omega$$

When $(\bar{T}_i - \bar{T}_j) \pm \omega$ covers 0



- Case 1: $\bar{T}_i - \bar{T}_j > 0$

$$\bar{T}_i - \bar{T}_j - \omega < 0 \Longrightarrow \bar{T}_i - \bar{T}_j < \omega$$

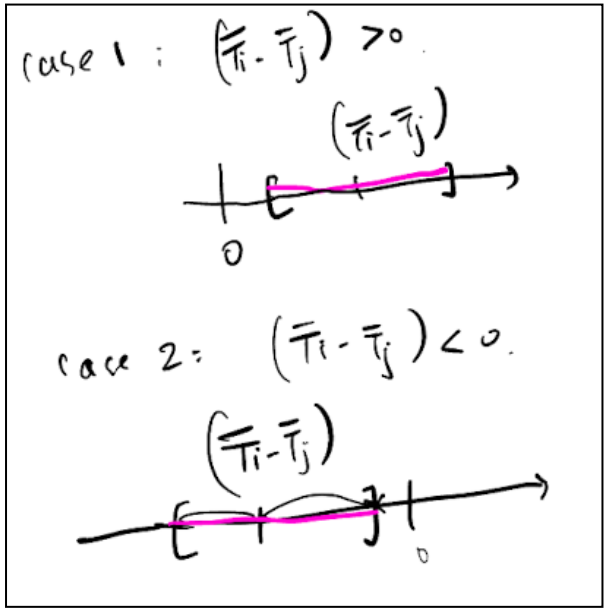
- Case 2: $\bar{T}_i - \bar{T}_j < 0$

$$\bar{T}_i - \bar{T}_j + \omega > 0 \Longrightarrow \bar{T}_j - \bar{T}_i < \omega$$

Both cases can be written as:

$$|\bar{T}_i - \bar{T}_j| < \omega$$

When $(\bar{T}_i - \bar{T}_j) \pm \omega$ does not cover 0



- Case 1: $\bar{T}_i - \bar{T}_j > 0$

$$\bar{T}_i - \bar{T}_j - \omega > 0 \Longrightarrow \bar{T}_i - \bar{T}_j > \omega$$

- Case 2: $\bar{T}_i - \bar{T}_j < 0$

$$\bar{T}_i - \bar{T}_j + \omega < 0 \Longrightarrow \bar{T}_j - \bar{T}_i > \omega$$

Both cases can be simplied as: $|\bar{T}_i - \bar{T}_j| > \omega$

A simple Rule to determine if treatment means are equal using Tukey's method

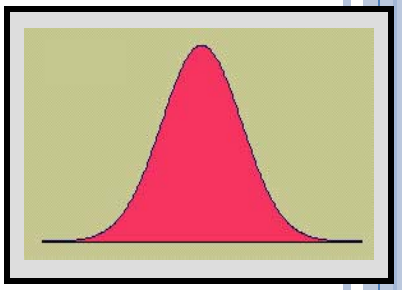
1. Calculate $\omega = q_\alpha(k, df)\dfrac{s}{\sqrt{n_i}}$
2. Rank $\bar{T}_i$'s from smallest to the largest
3. Calculate pairwise difference between $\bar{T}_i$ and $\bar{T}_j$.
4. If $|\bar{T}_i - \bar{T}_j| > \omega$, we conclude that the populaion means, μ_i and μ_j, are different.

NOTE

- If n_i are not equal, need to modify the Tukey's method (beyond the scope of this textbook).

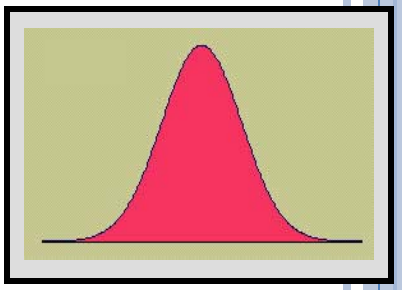
11.4 The randomized block design: a two-way classification

The Randomized Block Design



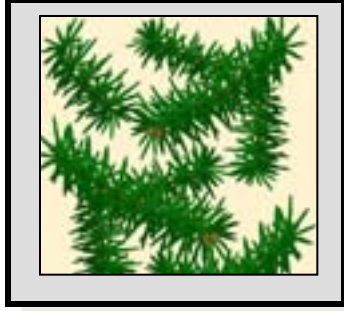
- A direct extension of the paired difference or matched pairs design.
- A **two-way classification** in which k treatment means are compared.
- The design uses **blocks** of k experimental units that are relatively similar or homogeneous, with one unit within each block randomly assigned to each treatment.

The Randomized Block Design



- If the design involves ***k* treatments** within each of ***b* blocks**, then the total number of observations is $n = bk$.
- The purpose of blocking is to remove or isolate the block-to-block variability that might hide the effect of the treatments.
- There are two factors—**treatments** and **blocks**, only one of which is of interest to the expeirmenter.

EXAMPLE



We want to investigate the affect of 3 methods of soil preparation on the growth of seedlings. Each method is applied to seedlings growing at each of 4 locations and the average first year growth is recorded.

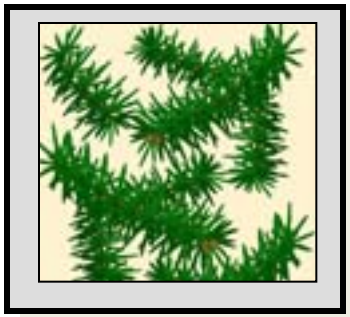
	Location			
Soil Prep	**1**	**2**	**3**	**4**
A	11	13	16	10
B	15	17	20	12
C	10	15	13	10

Treatment = soil preparation (k = 3)

Block = location (b = 4)

Is the average growth different for the 3 soil preps?

The Randomized Block Design



- Let x_{ij} be the response for the i-th treatment applied to the j-th block.
  - $i = 1, 2, \ldots k \qquad j = 1, 2, \ldots, b$
- The total variation in the experiment is measured by the **total sum of squares**:

$$\text{Total SS} = \sum (x_{ij} - \bar{x})^2$$

The Analysis of Variance

The **Total SS** is divided into 3 parts:

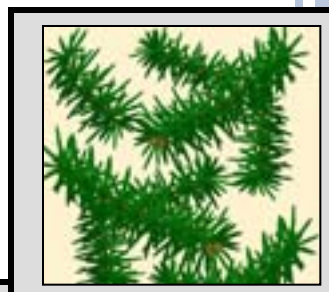
- ✓ **SST** (sum of squares for treatments): measures the variation among the k treatment means
- ✓ **SSB** (sum of squares for blocks): measures the variation among the b block means
- ✓ **SSE** (sum of squares for error): measures the random variation or experimental error

in such a way that:

$$\text{Total SS} = \text{SST} + \text{SSB} + \text{SSE}$$

Why?

Computing Formulas



$$\text{CM} = \frac{\text{G}^2}{n} \quad \text{where } \text{G} = \sum x_{ij}$$

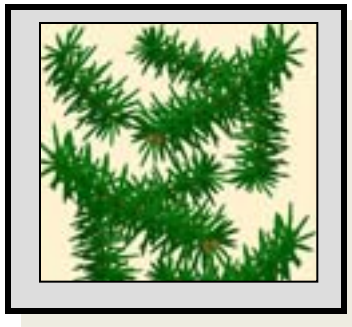
$$\text{Total SS} = \sum x_{ij}^2 - \text{CM}$$

$$\text{SST} = \frac{\sum T_i^2}{b} - \text{CM} \quad \text{where } T_i = \text{total for treatment } i$$

$$\text{SSB} = \frac{\sum B_j^2}{k} - \text{CM} \quad \text{where } B_j = \text{total for block } j$$

$$\text{SSE} = \text{Total SS} - \text{SST} - \text{SSB}$$

The Seedling Problem



	Locations				
Soil Prep	**1**	**2**	**3**	**4**	T_i
A	11	13	16	10	50
B	15	17	20	12	64
C	10	15	13	10	48
B_j	36	45	49	32	162

$$\text{CM} = \frac{162^2}{12} = 2187$$

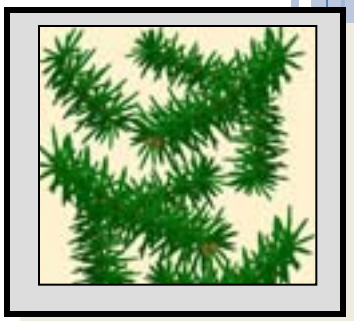
$$\text{Total SS} = 11^2 + 15^2 + \ldots + 10^2 - 2187 = 111$$

$$\text{SST} = \frac{50^2 + 64^2 + 48^2}{4} - 2187 = 38$$

$$\text{SSB} = \frac{36^2 + 45^2 + 49^2 + 32^2}{3} - 2187 = 61.6667$$

$$\text{SSE} = 111 - 38 - 61.6667 = 11.3333$$

The ANOVA Table



Total df = $bk - 1 = n - 1$

Treatment df = $k - 1$

Block df = $b - 1$

Error df = $(bk-1) - (k-1) - (b-1) = (k-1)(b-1)$

Mean Squares

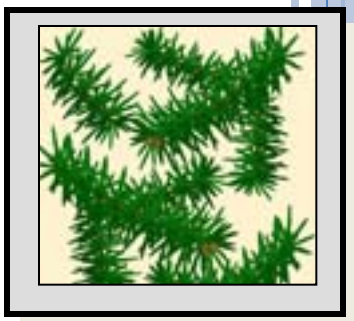
$MST = SST/(k-1)$

$MSB = SSB/(b-1)$

$MSE = SSE/(k-1)(b-1)$

Source	df	SS	MS	F
Treatments	k -1	SST	MST=SST/(k-1)	MST/MSE
Blocks	b -1	SSB	MSB=SSB/(b-1)	MSB/MSE
Error	(b-1)(k-1)	SSE	MSE=SSE/(b-1)(k-1)	
Total	n -1	Total SS		

The Seedling Problem



$$CM = \frac{162^2}{12} = 2187$$

$$\text{Total SS} = 11^2 + 15^2 + \ldots + 10^2 - 2187 = 111$$

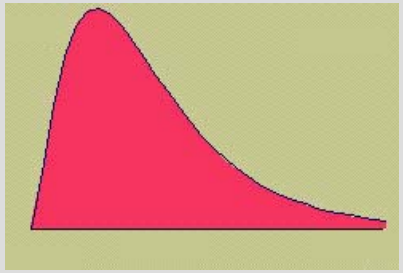
$$SST = \frac{50^2 + 64^2 + 48^2}{4} - 2187 = 38$$

$$SSB = \frac{36^2 + 45^2 + 49^2 + 32^2}{3} - 2187 = 61.6667$$

$$SSE = 111 - 38 - 61.6667 = 11.3333$$

Source	df	SS	MS	F
Treatments	2	38	19	10.06
Blocks	3	61.6667	20.5556	10.88
Error	6	11.3333	1.8889	
Total	11	122.9167		

Testing the Treatment and Block Means



For either treatment or block means, we can test:

$H_0 : \mu_1 = \mu_2 = \mu_3 = \ldots$ versus

H_a : at least one mean is different

Remember that σ^2 is the common variance for all bk treatment/block combinations. MSE is the best estimate of σ^2, whether or not H_0 is true.

- If H_0 is false and the population means are different, then MST or MSB— whichever you are testing— will unusually **large.** The test statistic **F = MST/ MSE** (or **F = MSB/ MSE**) tends to be larger that usual.
- We use a right-tailed F test with the appropriate degrees of freedom.

To test H_0 : treatment (or block) means are equal

$$\text{Test Statistic}: F = \frac{MST}{MSE} \left(\text{or } F = \frac{MSB}{MSE}\right)$$

Reject H_0 if $F > F_\alpha$ with $k-1$ (or $b-1$) and $(b-1)(k-1)$ df.

The Seedling Problem

Source	df	SS	MS	F
Soil Prep (Trts)	2	38	19	10.06
Location (Blocks)	3	61.6667	20.5556	10.88
Error	6	11.3333	1.8889	
Total	11	122.9167		

MY APPLET

df_2	a	1	2
1	.100	39.86	49.50
	.050	161.4	199.5
	.025	647.8	799.5
	.010	4052	4999.5
	.005	16211	20000
2	.100	8.53	9.00
	.050	18.51	19.00
	.025	38.51	39.00
	.010	98.50	99.00
	.005	198.5	199.0
3	.100	5.54	5.46
	.050	10.13	9.55
	.025	17.44	16.04
	.010	34.12	30.82
	.005	55.55	49.80
4	.100	4.54	4.32
	.050	7.71	6.94
	.025	12.22	10.65
	.010	21.20	18.00
	.005	31.33	26.28
5	.100	4.06	3.78
	.050	6.61	5.79
	.025	10.01	8.43
	.010	16.26	13.27
	.005	22.78	18.31
6	.100	3.78	3.46
	.050	5.99	5.14
	.025	8.81	7.26
	.010	13.75	10.92
	.005	18.63	14.54

To test for a difference due to soil preparation :

$H_0 : \mu_1 = \mu_2 = \mu_3$ versus

H_a : at least one mean is different

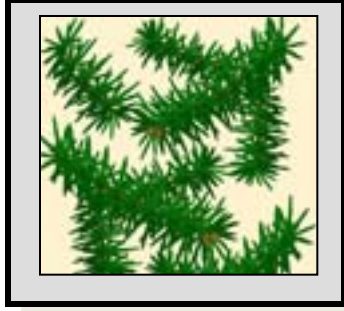
$$F = \frac{MST}{MSE} = 10.06$$

Rejection region : $F > F_{.05} = 5.14$.

We reject H_0 and conclude that there is a difference due to soil preparation.

Although not of primary importance, notice that the blocks (locations) were also significantly different (F = 10.88)

RECALL



We want to investigate the affect of 3 methods of soil preparation on the growth of seedlings. Each method is applied to seedlings growing at each of 4 locations and the average first year growth is recorded.

	Location			
Soil Prep	**1**	**2**	**3**	**4**
A (no prep)	11	13	16	10
B (fertilization)	15	17	20	12
C (burning)	10	15	13	10

Treatment = soil preparation (k = 3)

Block = location (b = 4)

Is the average growth different for the 3 soil preps?

Differences in treatment or block means?

If a difference exists between the treatment means or block means, we can explore it with

- Confidence intervals, or
- Using Tukey's method.

Confidence Intervals

Note $\left(\bar{T}_i - \bar{T}_j\right) \sim N\left(\mu_i - \mu_j, \frac{2\sigma^2}{b}\right) \Longrightarrow \frac{(\bar{T}_i - \bar{T}_j) - (\mu_i - \mu_j)}{\sqrt{2s^2/b}} \sim t_{df}$

Difference in treatments

$$(\bar{T}_i - \bar{T}_j) \pm t_{df;\alpha/2}\sqrt{s^2\left(\frac{2}{b}\right)}$$

where $\bar{T}_i = T_i/b$ are the treatment means, $df = (b-1)(k-1)$, and $s = \sqrt{MSE}$

Confidence Intervals

Note $\left(\bar{B}_i - \bar{B}_j\right) \sim N\left(\mu_i^B - \mu_j^B, \frac{2\sigma^2}{k}\right) \Longrightarrow \frac{(\bar{B}_i - \bar{B}_j) - (\mu_i^B - \mu_j^B)}{\sqrt{2s^2/k}} \sim t_{df}$

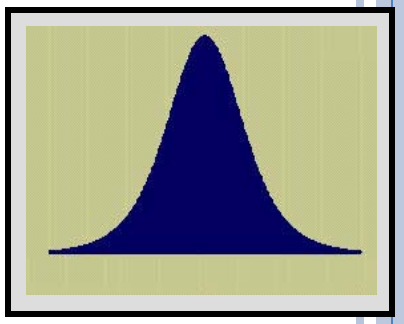
Difference in treatments $(\bar{B}_i - \bar{B}_j) \pm t_{df;\alpha/2}\sqrt{s^2\left(\frac{2}{k}\right)}$

where $\bar{B}_i = B_i/k$ are the treatment means, $df = (b-1)(k-1)$, and $s = \sqrt{MSE}$

Margin of error in Tukey's method

- $$\frac{(\bar{T}_{\max} - \bar{T}_{\min}) - (\mu_i - \mu_j)}{\frac{s}{\sqrt{b}}} \sim q(k, df)$$
- $$\frac{(\bar{B}_{\max} - \bar{B}_{\min}) - (\mu_i^B - \mu_j^B)}{\frac{s}{\sqrt{k}}} \sim q(b, df)$$
- $df = (b-1)(k-1)$

Tukey's Method



For comparing treatment means: $\omega = q_\alpha(k, df)\dfrac{s}{\sqrt{b}}$

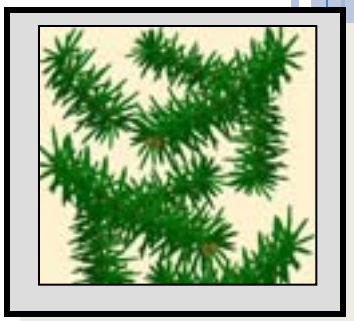
For comparing block means: $\omega = q_\alpha(b, df)\dfrac{s}{\sqrt{k}}$

$s = \sqrt{\text{MSE}}$ $\qquad df = \text{error } df$

$q_\alpha(k, df) = \text{value from Table 11.}$

If any pair of means differ by more than ω, they are declared different.

The Seedling Problem



Use Tukey's method to determine which of the three soil preparations differ from the others.

	A (no prep)	B (fertilization)	C (burning)
	$T_1 = 50$	$T_2 = 64$	$T_3 = 48$
Means	50/4 = 12.5	64/4 = 16	48/4 = 12

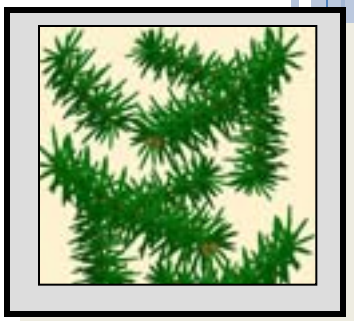
TABLE 11(a) Percentage Points of the Studentized Range, $q(k, df)$; Upper 5% Points

	k				
df	2	3	4	5	6
1	17.97	26.98	32.82	37.08	40.41
2	6.08	8.33	9.80	10.88	11.74
3	4.50	5.91	6.82	7.50	8.04
4	3.93	5.04	5.76	6.29	6.71
5	3.64	4.60	5.22	5.67	6.03
6	3.46	4.34	4.90	5.30	5.63
7	3.34	4.16	4.68	5.06	5.36
8	3.26	4.04	4.53	4.89	5.17
9	3.20	3.95	4.41	4.76	5.02

$$\omega = q_{.05}(3,6)\frac{s}{\sqrt{4}} = 4.34\frac{\sqrt{1.8889}}{\sqrt{4}} = 2.98$$

$$k = 3,\ df = (4-1)(3-1) = 6$$

The Seedling Problem



List the sample means from smallest to largest.

Soil Prep
A (no prep)
B (fertilization)
C (burning)

4.0

T_C	T_A	T_B
12	12.5	16.0

3.5

0.5

$$\omega = 2.98$$

Since the difference between 12 and 12.5 is less than ω = 2.98, there is no significant difference between population means C and A.

There is a difference between population means A and B.

There is a difference between population means C and B.

A significant difference in average growth only occurs when the soil has been fertilized.

Cautions about Blocking

- ✓ A randomized block design should not be used when treatments and blocks **both** correspond to experimental factors of interest to the researcher
- ✓ Remember that **blocking may not always be beneficial**.
- ✓ Remember that you **cannot construct confidence intervals for individual treatment means** unless it is reasonable to assume that the *b* blocks have been randomly selected from a population of blocks.